

Communication Skill for Software and Knowledge Engineers

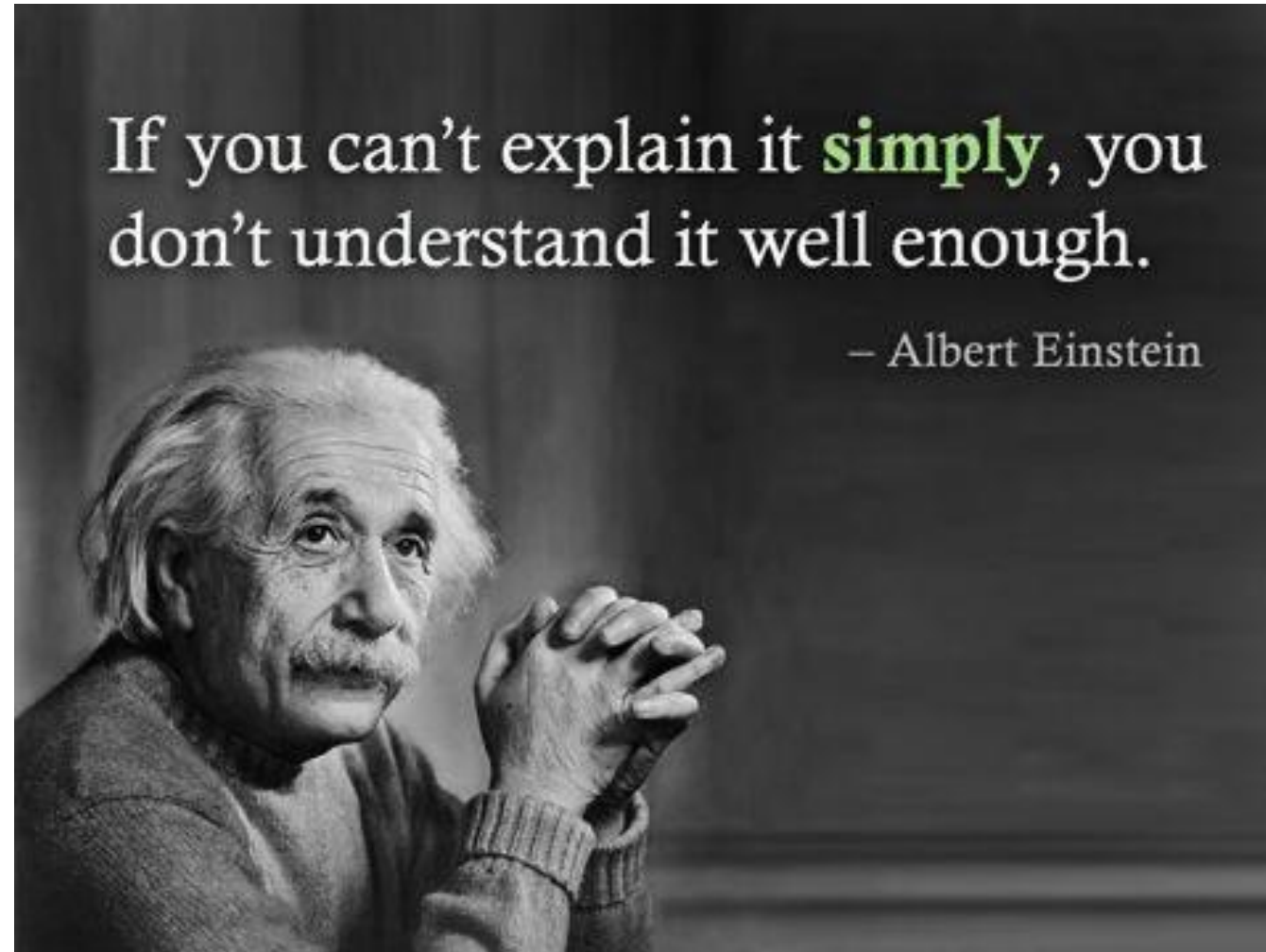
Lec. #04

Team Status Update Presentation

How to turn a research paper to presentation

Why Research Presentation Matter

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Why Research Presentation Matter

Research doesn't make an impact unless it's **understood**.

Key Reasons:

Communication = Impact

Even great research can be ignored if poorly presented.

Collaboration & Networking

Clear presentations attract the right collaborators, mentors, and funding.

Career Development

From academic conferences to job interviews, your ability to present well is a differentiator

Sharpen Your Thinking

Preparing to present forces you to **clarify ideas**, spot weak points, and organize logic.

Key Challenges Researchers Face

➤ Jargon Overload

- ✓ Hard for audiences to follow if terms aren't explained or simplified.

➤ Data Dumping

- ✓ Presenting **everything** leads to confusion. Focus is key.

➤ Stage Fright/Nerves

- ✓ Common even among experienced speakers. Can impact pacing, clarity, and confidence.

➤ Lack of Structure

- ✓ Ideas feel scattered when there's no clear beginning, middle, or end.

➤ Ineffective Slides

- ✓ Too much text, unreadable fonts, unclear charts.

Understand Your Audience

Why Audience Awareness is Critical

“If you don’t know who you’re talking to, you’re probably not reaching them.”

Tailoring your presentation to your audience ensures:

- Higher engagement
- Better retention of your key message
- Fewer misunderstandings or tune-outs

Understand Your Audience

Types of Audiences

Audience Type	Needs and Preferences	Presentation Tips
Experts in your field	Technical depth, precision, novelty	Go deeper into methods, show rigor
General academics	Clarity, big-picture relevance	Explain acronyms, simplify complex terms
Industry stakeholders	Practical implications, ROI, impact	Focus on results, scalability, applications
Public audience	Simplicity, stories, real-world relevance	Use analogies, avoid jargon, use visuals

Understand Your Audience

Types of Audiences

Audience Type

Experts in your field

General academics

Industry stakeholders

Public audience

- ✓ What does the audience **already know**?
- ✓ What do they **need to know to understand** my message?
- ✓ What are they **most interested in**?

Understand Your Audience

Tailor Effectively



Adjust depth and terminology

- Avoid jargon unless you define it clearly
- Replace formulas with intuition when appropriate



Adjust tone

- Academic tone: formal, structured
- Industry tone: concise, benefits-driven
- Public tone: storytelling, relatable



Use analogies and examples

- Translate technical concepts into familiar terms

Understand Your Audience Tailor Effectively



$$\overbrace{P(\theta|D, I)}^{\text{posterior}} \propto \overbrace{P(D|\theta, I)}^{\text{likelihood}} \overbrace{P(\theta|I)}^{\text{prior}}$$

Understand Your Audience Tailor Effectively

Bayesian inference

“Imagine updating your opinion based on new evidence, like changing your weekend plans depending on the weather forecast.”

Structure Your Presentation

“A good research talk is not a paper read aloud
— it’s a guided tour of your thinking.”

Why Structure Matters

Why Structure Matters

A well-structured talk:

- ❖ Keeps your audience engaged and oriented
- ❖ Helps you manage time and transitions
- ❖ Improves retention of your main ideas
- ❖ Makes complex content digestible

Key Components of a Research Talk Structure

1. Introduction (2–3 minutes)
2. Background & Motivation (3–5 minutes)
3. Methods (5–7 minutes)
4. Results (5–7 minutes)
5. Discussion & Implications (3–5 minutes)
6. Conclusion & Future Work (2–3 minutes)
7. Q&A (2–5 minutes)

Key Components of a Research Talk Structure

1. Introduction (2–3 minutes)

- ❑ Grab attention (hook, story, question, surprising fact)
- ❑ Clearly state your **research question**
- ❑ Briefly preview the roadmap of your talk

Use a “so what?” framing — why should the audience care?

Example:

“How can we predict earthquake damage using satellite images? This research explores a deep learning approach to map structural risk from space.”

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Key Components of a Research Talk Structure

2. Background & Motivation (3–5 minutes)

- ❑ Set the context

- ❑ What's the **real-world or scientific problem**?

- ❑ What have others done, and what's missing?

Avoid a literature dump; just enough to justify your contribution.

Example Slide: “Why current methods fail in post-disaster response”

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Key Components of a Research Talk Structure

3. Methods (5–7 minutes)

- ❑ Walk through what you did — clearly and logically
- ❑ Use diagrams, flowcharts, or step-by-step visuals
- ❑ Focus on your **main innovation or process**
- ❑ Skip technical derivations unless they are the contribution

Show the “how” only to the level your audience needs.

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Key Components of a Research Talk Structure

4. Results (5–7 minutes)

- ❑ Highlight **key findings**, not every number
- ❑ Use **clear graphs and figures** (not tables of numbers)
- ❑ Compare with baselines or expectations
- ❑ Highlight any surprising or notable outcomes

Tell a story with your results: “We expected X, but found Y.”

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Key Components of a Research Talk Structure

5. Discussion & Implications (3–5 minutes)

- ❑ What do your results **mean** in context?
- ❑ What are the **implications** for theory, application, or policy?
- ❑ Any unexpected observations or limitations?

Be honest and reflective. It builds credibility.

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Key Components of a Research Talk Structure

6. Conclusion & Future Work (2–3 minutes)

- ☐ Revisit your main message (don't assume it was absorbed)
- ☐ Briefly summarize your contribution
- ☐ Mention what's next — future directions, open questions

End with a clear **“so what”** message.

Key Components of a Research Talk Structure

7. Q&A (2–5 minutes)

- ☐ Budget time at the end for interaction
- ☐ Signal it clearly: **“Happy to take your questions now.”**
- ☐ Use backup slides if needed to support deeper questions

Pro Presentation Flow

**“Tell them what you’re going to say → say it
→ remind them what you said.”**

This reinforces your core message through
structure and repetition.

Designing Effective Slides

*“Slides should **support** your talk, **not replace** it.”*

Why Slide Design Matters

Why Slide Design Matters



Good

- Emphasize key ideas
- Guide attention
- Reinforce memory with visuals



Bad

- Confuse the audience
- Cause cognitive overload
- Undermine your credibility

Less Text, More Visuals

Do:

- Use *phrases* or *keywords*, not full sentences
- Replace text with figures, diagrams, flowcharts, icons, or images when possible
- Speak the details — the slide is a visual anchor

Don't:

- Copy-paste your paper
- Read off paragraphs
- Use full bullet lists for every slide



Font Sizes & Layout

- ❖ Minimum font size: **24pt**, ideally **28–32pt**
- ❖ Use **bold** or **color** to emphasize, not underline
- ❖ Keep layouts **clean**: lots of white space

Element	Recommended
Title	32–40 pt
Body text	28–32 pt
Labels/Legends	24 pt+



Graphs & Data Visualization

- ❖ Simplify the graph – **1 main idea per chart**
- ❖ Always **label axes, add units, and define terms**
- ❖ Avoid 3D effects or decorative chartjunk
- ❖ Use **consistent colors, legible fonts, and high contrast**

Example:

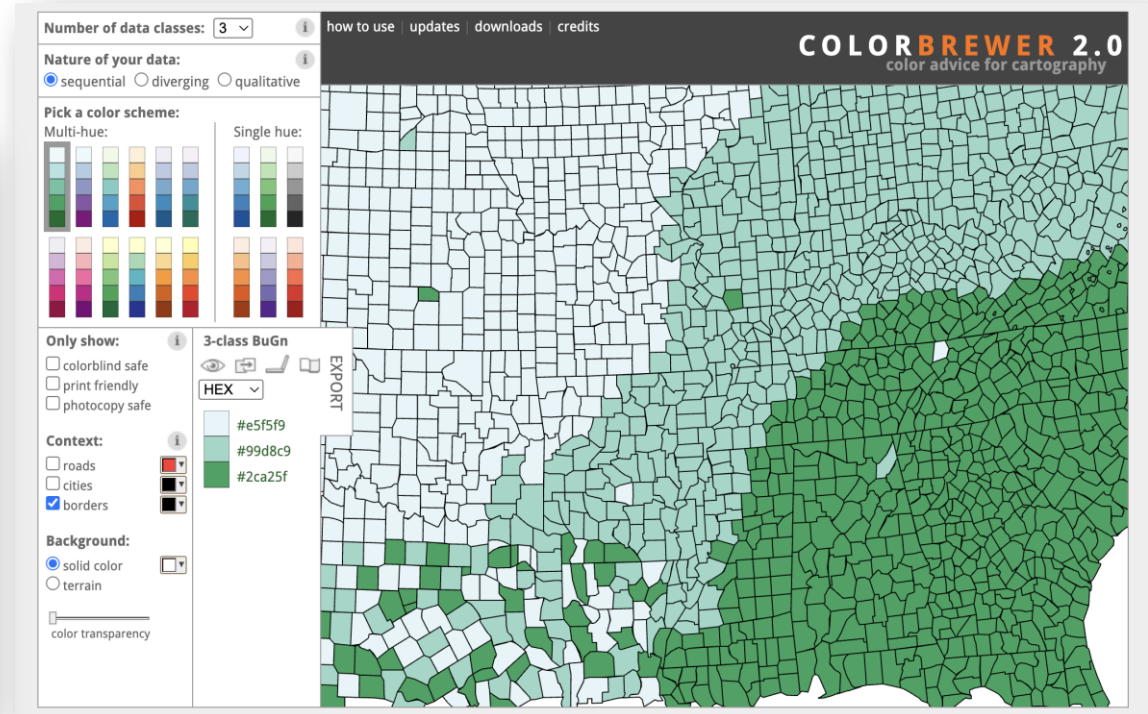
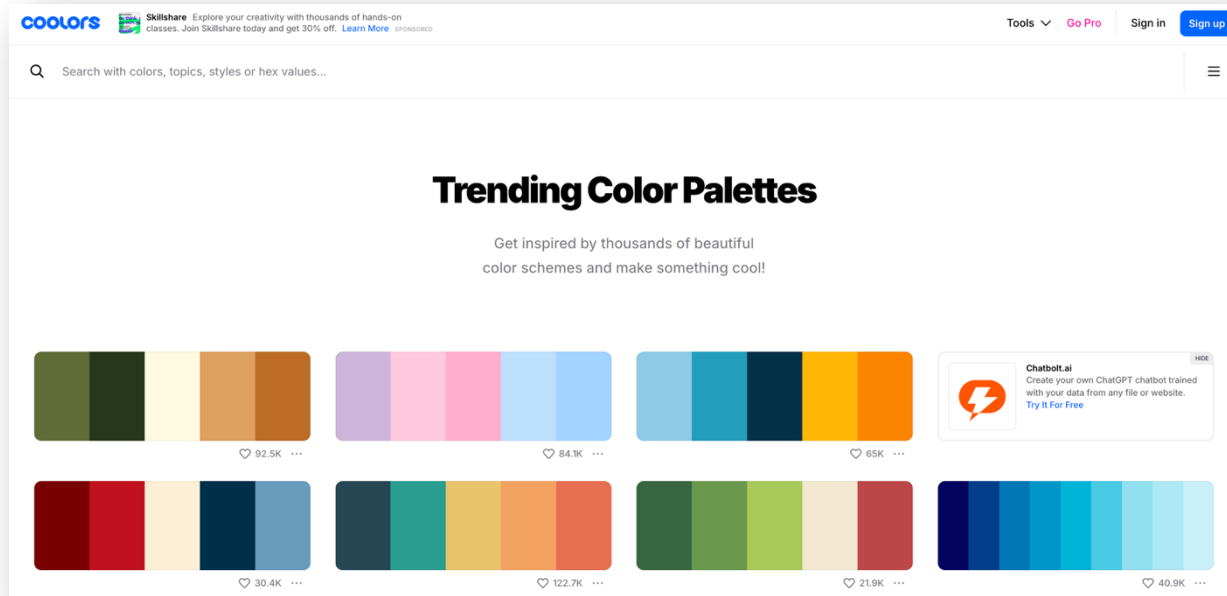
Bad: A cluttered matplotlib plot with 6 overlapping lines

Good: 1 line per slide, color-coded, with key takeaway highlighted



Color and Accessibility

- ❖ Use a **consistent color palette** (2–3 primary colors max)
- ❖ Avoid red/green for colorblind accessibility
- ❖ Use dark text on light background (or vice versa)
- ❖ Tools: ColorBrewer, Coolers





Avoid Overuse of Animation

- ❖ Use **simple transitions** (e.g., fade in)
- ❖ Avoid spinning, bouncing, or distracting effects
- ❖ Use animation **sparingly** to build one idea at a time
(e.g., highlight a point in a figure, reveal bullet points)



Highlighting Results Clearly

Show:

- ✓ Clear **before vs after** comparisons
- ✓ Use **boxes, arrows, or highlights** to guide attention
- ✓ Annotate surprising results or key differences
- ✓ Zoom in on areas of interest

Example Slide:

Left: Model prediction heatmap

Right: Ground truth → Highlight where they align / diverge



Highlighting Results Clearly

Label:

- ✓ Expected vs observed outcomes
- ✓ Confidence intervals
- ✓ Significance where relevant

Delivery Techniques

*“Great research isn’t enough
you have to make people care about it.”*





Voice: Your Most Underrated Tool

How you speak shapes how your message lands.

Vary Your:

- **Tone** – avoid monotone; express curiosity, urgency, or wonder
- **Pace** – slow down for key points, speed up for lighter transitions
- **Volume** – project clearly without shouting; use softness for emphasis
- **Pauses** – a few seconds of silence helps important points land

Pauses are powerful — use them instead of filler words ("um", "like").



Delivery Techniques

Eye Contact

- Scan the room — don't stare at one spot or the floor
- Make individual moments of connection with listeners

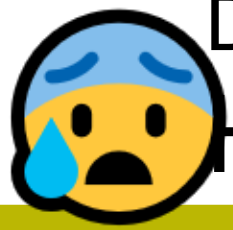


Delivery Techniques

Body Language

- Stand with good posture, feet planted
- Use open gestures to reinforce ideas
(e.g., “3 reasons” → show 3 fingers)
- Avoid nervous fidgeting (pen-clicking, pacing, crossing arms)

Imagine you're explaining your idea to a curious friend.



Delivery Techniques

Handling Nerves

Everyone gets nervous. What matters is how you manage it.

Techniques:

- **Box Breathing:** Inhale 4s → hold 4s → exhale 4s → hold 4s
- **Grounding:** Notice 3 things you see, 2 you hear, 1 you feel
- **Rehearse with intent:** Practice out loud, standing up, with slides
- **Positive framing:** You're sharing knowledge, not being judged

Confidence comes from preparation, not personality.

? Handling Q&A Effectively

“Q&A is not just a test - it’s an opportunity to connect, clarify, and demonstrate confidence.”

Why Q&A Matters

- ✓ Reinforces understanding of your topic
- ✓ Builds trust and engagement
- ✓ Reveals areas to improve or expand

? Handling Q&A Effectively

Step 1: Repeat the Question

Step 2: Answer Clearly & Concisely

Step 3: Handle What You Don't Know

Step 4: Diffuse Tough or Aggressive Questions

Step 5: Prepare in Advance

Step 1: Repeat the Question

Why?

- Ensures everyone hears it (especially in large or hybrid rooms)
- Buys you a few extra seconds to think
- Helps you **reframe** unclear or complex questions

Example:

“So you're asking whether this approach works for larger datasets — great, let's talk about that...”

Step 2:

Step 3:

Step 4:

Step 5:

Step 2: Answer Clearly & Concisely

- Stick to **one main point per answer**
- Use structure if helpful:
 - “Three reasons why that matters...”
- Tie it back to your research question or motivation
- If possible, **connect your answer to audience interests**

Avoid going into unrelated tangents — keep it focused.

Step 3:

Step 4:

Step 5:

Step 3: Handle What You Don't Know

It's okay to say “I don't know” — just do it with grace.

Examples:

- “That's an excellent point — I'll need to dig into the data to confirm.”
- “I haven't looked into that angle yet, but it's worth exploring.”
- “I don't have that number off-hand, but I'll follow up after the session.”

Have a slide or link ready for follow-up contact/info.

Step 4:

Step 5:

Step 4: Diffuse Tough or Aggressive Questions

Some questions aren't really questions — they're critiques in disguise.

Stay calm and professional:

- “Thanks for raising that — here’s how I see it.”
- “That’s a valid concern. Let me explain our decision logic.”
- “I appreciate your perspective — this is definitely a nuanced issue.”

Don't:

- Get defensive
- Rush to answer without understanding
- Take it personally

Step 5:

Step 5: Prepare in Advance

Make a short list of **expected or tricky questions**:

Type	Example
Clarification	“How is this different from X?”
Limitation	“What are the weaknesses of your method?”
Scope	“Can this be applied to real-time systems?”
Assumptions	“Why did you choose this model?”

Prepare **short, confident responses** with backup slides if helpful.

Example

Ref: <https://ieeexplore.ieee.org/document/10467799>

Proc. of International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA 2024)
1-2 February 2024, Victoria-Seychelles

A Quantum Annealing-Based Approach for Solving Talent Scheduling

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Abstract—Talent scheduling is a complicated task that arises from the intricate interplay of human resources, project dynamics, and organizational goals. The inherent complexities are magnified when managing tasks that involve a diverse array of factors, including skills, project timelines, and resource limitations. This paper introduces a novel approach employing quantum annealing to address the complex challenge of talent scheduling. Quantum annealing, an emerging field known for its innate optimization capabilities, offers a compelling framework for addressing this multifaceted challenge. We present a comprehensive exploration of our proposed quantum annealing-based method, comparing its efficacy against the traditional dynamic programming approach and various meta-heuristic algorithms. Our experimental results reveal that the proposed quantum annealing-based approach achieves optimality equivalent to the dynamic programming approach but with significantly reduced runtime. Additionally, we examine the outcomes when compared to other meta-heuristic approaches, shedding light on the comparative strengths and limitations. Furthermore, we investigate the impact of varying the number of samples and sweeps within our quantum annealing-based approach, providing insights into the sensitivity of our method to these parameters.

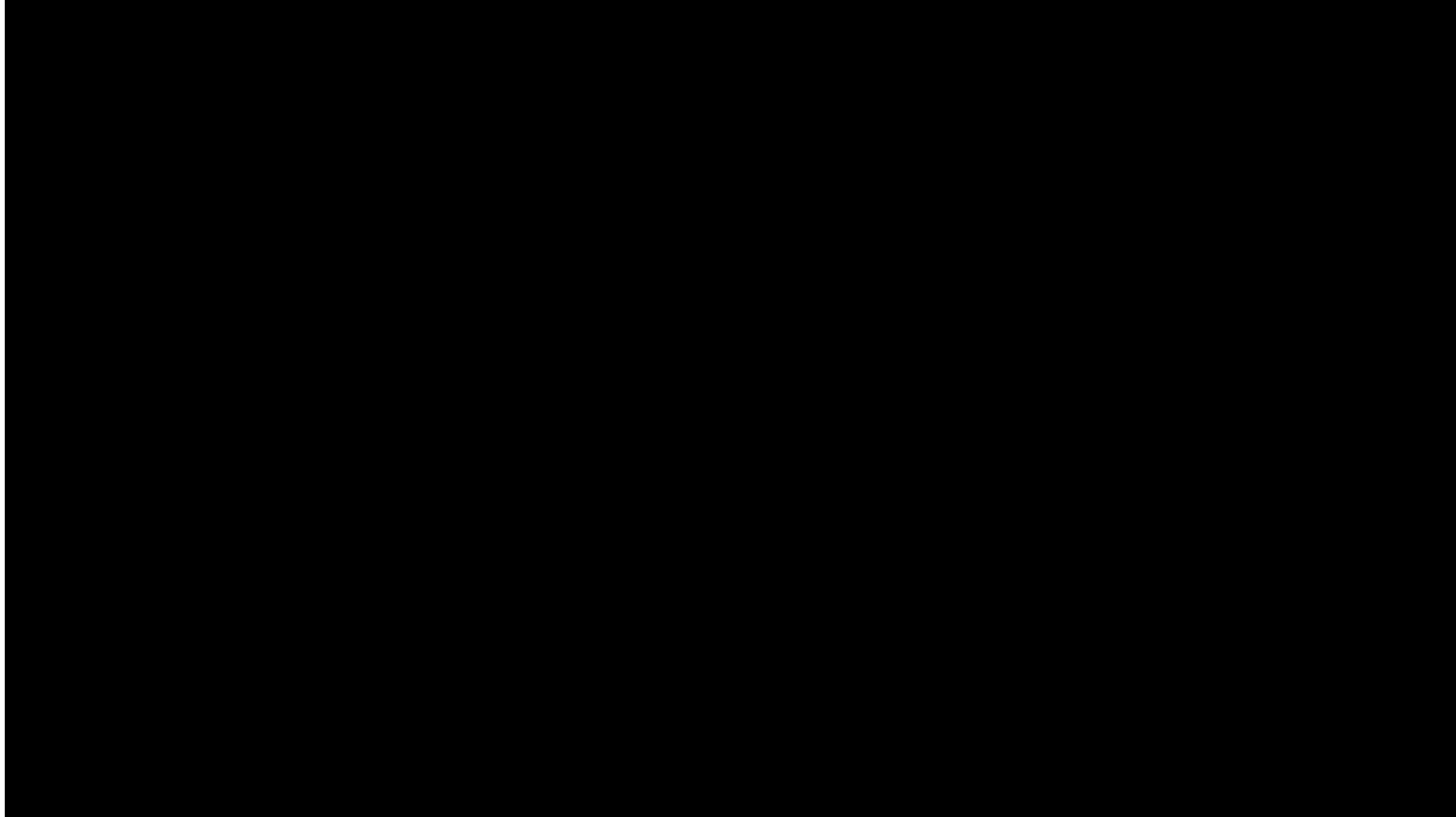
Index Terms—Combinatorial Optimization, Dynamic Programming, Metaheuristic Algorithm, Quantum Annealing, Talent Scheduling



Fig. 1. An example of total cost reduction in talent scheduling

which to shoot these scenes serves as a primary motivation for the exploration of this problem. Through the lens of a film producer responsible for orchestrating scene schedules at

Example



Video



Q&A

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